

[redacted] Functional Requirements

Created by Daugherty, Austin, last modified on Feb 05, 2024

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Document owner	@ Daugherty, Austin
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Introduction

This project involves the development of an **application** through which [redacted] members, [redacted] managers, and [redacted] teams can view and better understand the lineage of data as it flows through the [redacted]'s network. Currently, access to this lineage data is difficult to obtain and requires the user to go through many tedious and redundant steps. However, because this data is stored within the [redacted]'s network and is presumably accessible upon request, this means that we can pre-calculate lineage and stitch together a picture for users to access when required. This would reduce the need to trace lineage back manually as this could be seen in a single picture.

Our vision of a final product involves a **2D network visualization** that allows the user¹ to see how **data, applications, and reports** within the [redacted]'s network are **connected**. These pictures would be pre-calculated and available on demand making the process much smoother than was prior. The primary objective for the application would be to provide data understanding for a selected [redacted] or [redacted] Report. Therefore, the user will select a specific [redacted] or [redacted] before being presented with a subset of the [redacted]'s network that is associated with said entry. It is important to note that because the [redacted]'s network is a size beyond comprehension, this subset is required to provide the user with a smooth viewing experience.

Also, the user can further explore the selected subset by highlighting certain associations. Associations will include but not limited to:

- The path of a [redacted] through the network
- The path of all [redacted] **directly related to a selected** [redacted] through the network.
- The path of all data elements related to a **selected report** through the network.
- The path of all data elements related to all reports associated with a **selected regulator**.

At any point the user may reset the visualization to return to the original subset or select a new one.

1. See system roles below for more detail.

Roles:

- **User:** Any member of the [redacted] [redacted] Business, or Reporting teams that use the application to visualize data flow.

Requirements

1. The **system** shall prompt the user upon visiting the application to select an [redacted] or Report they would like to see a subset of the [redacted]'s network for.

#	User Story
1	The prompt may be switched between showing [REDACTED] loaded into the system or Reports loaded into the system.
2	The prompt shall be a dropdown list with a set number of entries to choose from
3	The prompt shall default to display [REDACTED] options
4	The prompt shall default to a blank value
5	The user may change the input at any time

2. The **system** shall display a network visualization modeling the [REDACTED]'s network of [REDACTED], [REDACTED], [REDACTED] and Other sources in a selected subset

#	User Story
1	The system may display up to 1 hops removed connections upstream from the source [REDACTED]
2	The system shall distinguish [REDACTED], [REDACTED], Reports, and Other sources from each other
3	The system shall distinguish [REDACTED] from [REDACTED]
4	The system shall distinguish [REDACTED] and Reports with [REDACTED] incidents from those that do not
5	The system shall differentiate network edges based on usage

3. A **user** may choose to interact with the model highlighting important information in the subset

#	User Story
1	The user may select a specific [REDACTED] to highlight its flow through the network
2	The user may select a specific Report to highlight its associated [REDACTED] through the network.
3	The user may select a specific Regulator to see [REDACTED] associated with all Reports linked to the selected Regulator and how they flow through the network.
4	The user may select a specific [REDACTED] to see how all [REDACTED]'s directly associated with the selected application flow through the network.
5	The user's selections from 3.1-3.4 may compound on top of each other.
6	The user may reset their selections at any time.
7	The user may hover over a Report node to see it's percent of data is coming from associated source [REDACTED](s) in the view].
8	The user may hover over an [REDACTED] to see its related data and up to 5 associated [REDACTED]
9	The user may hover over a [REDACTED] to see its related data
10	The user may hover over an 'Other' source to see more detail

4. The **system** may filter to a "system" mode.

#	User Story
1	The user may toggle between System and [REDACTED] mode.

2	The system in [REDACTED] mode displays the existing network
3	The system in system mode displays the existing network without associated [REDACTED]
4	The user's choice is persisted until toggled back to [REDACTED] mode

5. The **user** may export the visualization data

#	User Story
1	Data shall include mappings between Reports and their associated [REDACTED] and the [REDACTED] the [REDACTED] flow through
2	Data shall include metadata related to Reports in the view
3	Data shall reflect any filters selected by the user

Non-Functional Requirements:

- 1. No **user's** session should be persisted from one session to the next
- 2. No **user's** session should affect a different user's session.
- 3. The **system** shall be refreshed on a **weekly**
- 4. The **system's** extract files (report to source) shall be refreshed on a **quarterly**
- 5. The model should be pre-calculated for the user upon arrival.
- 6. The visualization should be intuitive and easy to use.
- 7. The application should be web hosted.

Constraints:

- 1. The network visualization model shall use force directed network principles.
- 2. Exported data shall be in a XLSX format.